Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

CRITICAL CARE

YG CG 1007 MAY 2021.

DIABETIC KETOACIDOSIS (DKA)

REVIEW BY (MAY 2024)

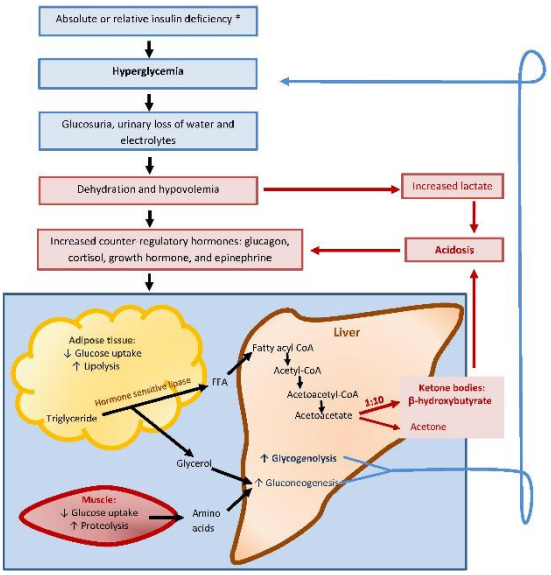
Author(s): Eirlys Walker
Reviewer: Anaesthetic Departmental Guideline Group

Key Points

- Complex disordered metabolic state
- **Should be treated as an emergency**
- Patients will likely present with severe fluid depletion and possible electrolyte disruption
- Resolution should be possible within 24 hours
- Involvement of Diabetic Specialist team to advise on ongoing insulin requirements post resolution
- To be read in conjunction with YG CG xxx "Diabetic Emergencies"

INTRODUCTION

Diabetic Ketoacidosis (DKA) is a complex disordered metabolic state that occurs as a result of an absolute or relative deficiency of insulin, accompanied by an increase in the counter regulatory hormones. The resultant hepatic glyconeogenesis and glycogenolysis and increased lipolysis and breakdown of the resulting free fatty acids as an alternative energy source, leads to the characteristic biochemical triad of ketonaemia (ketosis), hyperglycaemia and acidosis. DKA is usually associated with Type 1 diabetes, but increasingly there are cases of ketone prone Type 2 diabetics being seen. Incidence, although difficult to accurately measure, is believed to range from 4.6 to 8 episodes per 1000 patients with diabetes. Improved recognition, understanding and established treatment regimens have seen the mortality rates fall over the last 20 years from 7.6% to 0.67%. However, mortality rates remain high in developing countries and in out of hospital situations. The onset of DKA is fairly rapid – usually developing within 24 hours.



DIAGNOSIS

The presence of all three of the following:

- *Capillary blood glucose >11mmol/L or known Diabetes Mellitus*
- *Blood Ketones >3mmol/L or urine ketones ++or above*
- *Venous pH <7.3 &/or bicarbonate <15mmol/L*

The presence of one or more of the following may indicate severe DKA and need for level 2 care. CVC insertion and senior review should be considered:

Blood ketones >6mmol/L *venous bicarb <5mmol/L* *venous pH <7.1* *Anion gap >16*
Hypokalaemia on admission (< 3.5) *O2 Sats <92% on air (assuming normal resp func)*
Systolic BP <90mmHg *HR >100 or <60* *GCS <12*

MANAGEMENT

Management is split into 4 main time frames:

A slower rate &/or reduced volume of fluid should be considered if patients are between 18 and 25yrs, >70yrs, pregnant or have existing renal or cardiac failure.

IMMEDIATE MANAGEMENT UPON DIAGNOSIS (0-60 MINUTES)	SUBSEQUENT MANAGEMENT (60 MINS – 6HRS)	At 6 – 12 HRS	At 12 – 24 HRS
T=0 at time IV fluids started ACTION 1: FLUIDS Start IV fluids 0.9% Sodium Chloride as per prescription chart Consider giving a rapid initial 500ml bolus if SPB <90mmHg ACTION 2: INSULIN Commence fixed rate insulin infusion - 0.1 unit/Kg/Hr (If weight unknown then estimate) Max recommended rate 15units/hr - Ensure any long-acting insulin that the patient usually takes is prescribed at normal dose and time ACTION 3: ASSESSMENT - RR, HR,BP,Temp, O2 sats - GCS - Full clinical examination ACTION 4: FURTHER I* To include FBC, U&Es, Capillary and lab glucose, VBG, blood ketones, blood cultures, Urinalysis and urine C&S, CXR and ECG ACTION 5: MONITORING Hourly glucose and ketones; Repeat U&Es at 60min Consider precipitating factors and treat. Consider thromboprophylaxis	AIMS OF TREATMENT: - Rate of fall of Ketones of at least 0.5mmol/L/Hr OR Bicarbonate rise of 3mmol/L/Hr - Fall in Blood glucose of 3mmol/L/Hr ACTION 1: REASSESS PATIENT: - Hrly capillary blood glucose and ketones - VBG for pH, bicarb and K+ at 60mins, 2hrs and 2hrly thereafter Follow guidance for potassium replacement on prescription portion of chart (if K+ outside normal range reassess replacement, recheck hourly and consider senior review Complete monitoring portion of form ACTION 2: CONTINUE IV FLUID REPLACEMENT: Follow the fluid schedule on the chart When blood glucose is ,14mmol/L add in 10% Glucose infusion to run concurrently *DO NOT STOP INSULIN* If above treatment aims not being met check infusion pump and IV access are okay. If okay consider increasing insulin infusion TARGET: Ketones ,3mmol/L Venous pH >7.3 &/or Bicarb >18mmol/L Consider catheter if anuric or incontinent. Consider NG if vomiting *Ensure referral to Diabetes Team	ACTION 1: CONTINUE IV FLUID REPLACEMENT - Follow fluid replacement schedule on prescription portion of chart - Assess for complications of treatment e.g. fluid overload or cerebral oedema *Any sudden deterioration in GCS should be considered likely cerebral oedema unless definitively proven otherwise* ACTION 2: REVIEW BIOCHEMISTRY/INSULIN INFUSION - At 6hrs check venous pH, potassium and bicarb and check capillary glucose and ketones - Avoid hypoglycaemia - Ensure continued biochemical and clinical improvement. - Reassess patient and monitor vital signs - If patient not improving as per aims in Box 2 seek senior review DEFINITION OF RESOLUTION: - Ketones <0.3mmol/L - pH >7.3 (Do not use bicarb at this stage) If DKA resolved: switch to VRII as per local guidelines until patient eating and drinking If DKA not resolved: review insulin infusion as per Box 2	By 24 hours ketonaemia and acidosis should have resolved If not improving seek senior review and review insulin and fluids as per Box 2 At 12 hrs check Venous pH, K+ and bicarb and check capillary Glucose and ketones Aims: - Ensure clinical and biochemical markers continue to improve - Continue iV fluids if not eating and drinking - Reassess for treatment complications If ketonaemia resolved and patient is NOT eating and drinking: move to VRII as per local guidelines If ketonaemia resolved and the patient IS eating and drinking normally: move to SC insulin and stop IV fluids *DO NOT DISCONTINUE IV INSULIN INFUSION UNTIL 30 MINS AFTER SHORT ACTING SC INSULIN HAS BEEN GIVEN* *DIABETES TEAM REVIEW PRIOR TO DISCHARGE*

References

Join British Diabetes Societies Inpatient Care Group. *Management of Diabetic Ketoacidosis in Adults 2nd Edition;* Sept 2013

BCUHB Adult Diabetic Ketoacidosis Prescription Chart
http://howis.wales.nhs.uk/sitesplus/documents/861/2020%20BCUHB%20DKA%20Guideline%20Version%202%20approved%20DTG%20Final.pdf